7	**(a)**	e.g. ‘instantaneous’ emission (of electrons) threshold frequency below which no emission (max) <u>electron</u> energy dependent on frequency (max) <u>electron</u> energy not dependent on intensity rate of emission (of electrons) depends on intensity (*any three sensible suggestions, 1 each*)		B3	[3]
	(b)	**(i)**	‘packet’ / quantum of energy of electromagnetic energy / radiation	M1 A1	[2]
		(ii)	discrete wavelengths mean photons have particular energies energy of photon determined by energy change of (orbital) electron so discrete energy levels	M1 M1 A0	[2]
	(c)	**(i)**	three energy changes shown correctly arrows ‘pointing’ in correct direction wavelengths correctly identified	B1 B1 B1	[3]
		(ii)	chooses λ = 486 nm $\Delta E = hc / \lambda$ $= (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (4.86 \times 10^{-9})$ $= 4.09 \times 10^{-19}$ J (*allow 2 s.f.*)	C1 C1 A1	[3]
8	**(a)**	region (of space) / area where		B1	